

MODULE 1

MULTIMEDIA BASICS

Introduction

Multimedia is an interactive media and provides multiple ways to represent information to the user in a powerful manner. It provides an interaction between users and digital information. It is a medium of communication. Some of the sectors where multimedia's is used extensively are education, training, reference material, business presentations, advertising and documentaries.

Definition of Multimedia

By definition Multimedia is a representation of information in an attractive and interactive manner with the use of a combination of text, audio, video, graphics and animation. In other words we can say that Multimedia is a computerized method of presenting information combining textual data, audio, visuals (video), graphics and animations. For examples: E-Mail, Yahoo Messenger, Video Conferencing, and Multimedia Message Service (MMS).

Multimedia involves multiple modalities of text, audio, images, drawings, animation, and video.

Examples of how these modalities are put to use:

1. Video teleconferencing.
2. Distributed lectures for higher education.
3. Tele-medicine.
4. Co-operative work environments.
5. Searching in (very) large video and image databases for target visual objects.
6. "Augmented" reality: placing real-appearing computer graphics and video objects into scenes.
7. Including audio cues for where video-conference participants are located.
8. Building searchable features into new video, and enabling very high- to very low-bit-rate use of new, scalable multimedia products.
9. Making multimedia components editable.
10. Building "inverse-Hollywood" applications that can recreate the process by which a video was made.
11. Using voice-recognition to build an interactive environment, say a kitchen-wall web browser.

Multimedia and Hypermedia

Multimedia:

Multimedia refers to the integration of multiple types of media to convey information, including text, audio, graphics, images, video, and animation. It allows computer systems to present and process data through a variety of media formats. Multimedia focuses on delivering information using both traditional text and graphic elements combined with richer forms like audio and video.

Hypermedia:

Hypermedia extends the concept of hypertext, which was introduced by Ted Nelson around 1965. While hypertext allows non-linear navigation of text documents, hypermedia includes other media types such as graphics, images, audio, and video. Hypermedia is not limited to text-based content; it incorporates various media to create interconnected, non-linear navigation systems.

The World Wide Web (WWW) is a prime example of a hypermedia application.

In essence, hypermedia is a specialized form of multimedia with a focus on linking various media elements in a non-linear, interactive way.

Hypertext

Definition: Hypertext is a nonlinear way of presenting information. Unlike normal text, which is read sequentially (linear), hypertext allows users to navigate information freely through "hot spots" (links) that connect to other related pieces of information.

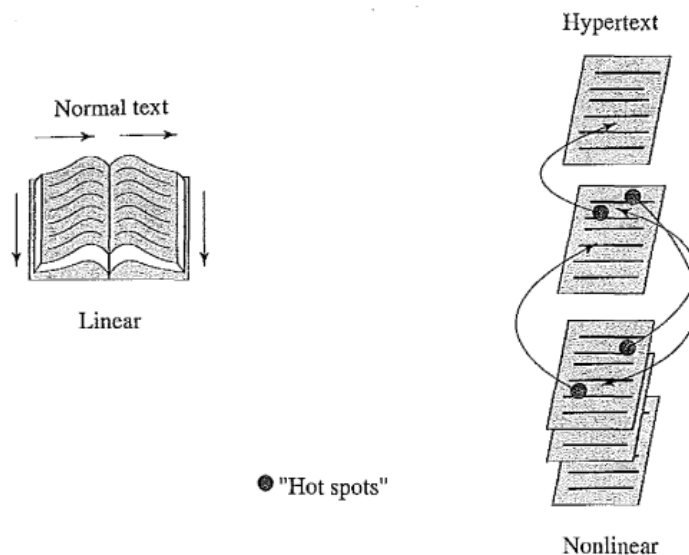


FIGURE 1.1: Hypertext is nonlinear.

Depicts the difference between normal text (linear, read from start to end) and hypertext (nonlinear, navigable through links or hotspots). Hotspots provide clickable areas to access other parts of the document or other resources.

Multimedia Applications

Digital video editing and production systems

Electronic newspapers and magazines

The World Wide Web (WWW)

Online reference works (encyclopedias)

Games, groupware, interactive TV

Multimedia courseware, video conferencing, and video-on-demand

World Wide Web (WWW)

Definition: The WWW is a hypermedia application that became the most popular and widely used networked system for accessing and sharing information.

Features of its popularity:

1. Amount of information: Large databases available from web servers.
2. Ease of navigation: Hypermedia links and web browsers make navigation simple.
3. Maintenance: Developed and maintained by organizations like the W3C (World Wide Web Consortium).

- **HTTP (HyperText Transfer Protocol)**

Definition: HTTP is the foundation of data communication for the World Wide Web. It is a protocol used for transmitting hypermedia documents, such as HTML.

The basic request format is

```
Method URI Version
Additional-Headers
```

```
Message-body
```

The basic response format is

```
Version Status-Code Status-Phrase
Additional-Headers
```

```
Message-body
```

Key Features:

1. Request-Response Protocol: HTTP operates on a client-server model.
A client (browser) sends a request to a web server, and the server responds with the requested resource (e.g., a web page).
2. Methods:
GET: Used to retrieve data from a server. Commonly used for fetching web pages.
POST: Used to send data to a server, such as form submissions.
3. Uniform Resource Identifier (URI):
Identifies the resource being accessed. Usually starts with http:// or https:// (secure HTTP).
Includes the host name and the path to the resource.
4. Headers and Body:
Headers: Provide additional information about the request/response, such as content type and status codes.
Message Body: Contains the actual data being sent (e.g., form data in a POST request).
5. Stateless Protocol:
Each HTTP request is independent and does not retain information about previous requests.
Example: A URL like http://www.example.com identifies the server (www.example.com) and the resource to be fetched.
A GET request to the server fetches the resource, while a POST request can upload data to it.

- **HTML (HyperText Markup Language)**

HTML (HyperText Markup Language) is the standard language used to create and design web pages. It structures web content using a system of elements and tags that define how text, images, links, and other media appear on a webpage.

HTML document is structured using various elements and tags that define its content and appearance. Below is the basic structure of an HTML document:

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>
  <h1>Main Heading</h1>
  <p>This is a paragraph.</p>
</body>
</html>
```

simple example of an HTML document:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First HTML Page</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This is a paragraph of text.</p>
  <a href="https://www.example.com">Visit Example</a>
</body>
</html>
```

- **XML**

XML (Extensible Markup Language) is a markup language designed to store, transport, and organize data in a structured way. It is both human-readable and machine-readable, making it widely used for data exchange between different systems, especially over the internet.

Key Features of XML

Self-Descriptive – XML allows users to define their own tags (e.g., <book>, <price>).

Hierarchical Structure – Data is stored in a tree-like format with parent and child elements.

Platform-Independent – Works across different systems and programming languages.

Strict Syntax Rules – Tags must be properly nested and closed.

Supports Unicode – Can store text in any language.

Basic XML Example

xml

Copy code

```
<?xml version="1.0" encoding="UTF-8"?>
<library>
  <book>
    <title>XML Basics</title>
```

```

    <author>John Doe</author>
    <year>2023</year>
    <price>29.99</price>
  </book>
</book>
  <title>Advanced XML</title>
  <author>Jane Smith</author>
  <year>2024</year>
  <price>39.99</price>
</book>
</library>

```

Explanation:

Root element: <library> contains all data.

Child elements: <book> represents individual items.

Sub-elements: <title>, <author>, <year>, and <price> store details.

Why Use XML?

- ✓ Data Exchange: Used in web services (SOAP, REST APIs).
- ✓ Configuration Files: Found in software settings (.xml files).
- ✓ Database Storage: Used for structured data representation.
- ✓ Interoperability: Works across different platforms and applications.

Xml specifications

1.XML Protocol:- Used to exchange XML information between processes. It is meant to supersede HTTP and extend it as well as to allow interprocess communications across networks.

2. XML Schema:- A more structured and powerful language for defining XML data types (tags). Unlike a DTD, XML Schema uses XML tags for type definitions.

3. XSL:- This is basically CSS for XML. On the other hand, XSL is much more complex, having three parts: XSL Transformations (XSLT), XML Path Language (XPath), and XSL Formatting Objects.

- **SMIL (Synchronized Multimedia Integration Language)**

SMIL (Synchronized Multimedia Integration Language) is a markup language designed for publishing multimedia presentations, similar to how HTML is used for text-based documents. It allows for the scheduling and synchronization of various multimedia elements, such as text, audio, video, and images, enabling them to change dynamically over time. Unlike text documents, which are displayed sequentially, SMIL presentations involve multiple elements interacting with users while playing simultaneously or in a coordinated manner.

Key Features of SMIL:

Multimedia Synchronization: SMIL allows precise control over the timing of media elements, ensuring that videos, audio, and text appear at the right moments.

Scheduling & Layout Control: Authors can specify when and where different media elements should be displayed on the screen.

Interactivity: SMIL enables user interaction by defining clickable links, buttons, and other interactive elements within multimedia presentations.

Lightweight & Web-Friendly: Unlike heavy multimedia applications, SMIL is a lightweight alternative that can be used in web browsers and multimedia players.

Text-Based & Easy to Edit: Since SMIL is XML-based, it can be written and edited using any text editor or XML editor.

Uses of SMIL:

Online training courses and educational content

Multimedia presentations and slideshows

Digital storytelling

Interactive web-based advertisements

Subtitling and captioning in videos

Example:

```
<smil>
  <head>
    <layout>
      <region id="videoRegion" top="10" left="10" width="640" height="360"/>
      <region id="textRegion" top="380" left="10" width="640" height="100"/>
    </layout>
  </head>

  <body>
    <seq>
      <!-- Display a video -->
      <video src="video.mp4" region="videoRegion" dur="10s"/>

      <!-- Play an audio file in sync with the video -->
      <audio src="audio.mp3" dur="10s"/>

      <!-- Show a text caption after the video starts -->
      <text src="caption.txt" region="textRegion" begin="2s" dur="6s"/>
    </seq>
  </body>
</smil>
```

- The categories of software tools:-

1. Music sequencing and notation
2. Digital audio
3. Graphics and image editing
4. Video editing
5. Animation
6. Multimedia authoring.

Music Sequencing and Notation Tools

These tools are used for music composition, sequencing, and editing.

Cakewalk: A well-known MIDI sequencing and editing software. It helps musicians store sequences of notes, insert WAV files, and even add animations and video to music tracks.

Cubase: A music sequencing and editing program similar to Cakewalk. It includes features for digital audio editing and MIDI composition.

Digital Audio Tools

These tools are used for recording, editing, and processing audio.

Cool Edit: A powerful digital audio toolkit that simulates a professional audio studio. It supports multitrack production and digital signal processing effects.

Sound Forge: A PC-based program for editing WAV files. It allows audio capture from CD-ROM drives, tapes, and microphones, and includes special effects.

Pro Tools: A high-end audio production and editing software for Mac and Windows. It supports MIDI creation, audio mixing, recording, and editing

Graphics and Image Editing Tools

These tools are used for creating, editing, and manipulating images and graphics.

Adobe Illustrator: A professional tool for vector graphics editing that is widely used for web design and print media.

Adobe Photoshop: The industry-standard software for image processing, graphics design, and photo editing. It supports layers, filters, and advanced image effects.

Macromedia Fireworks: A tool designed specifically for web graphics. It includes a bitmap editor, a vector graphics editor, and a JavaScript generator for interactive elements.

Macromedia Freehand: A vector-based graphics tool used for creating text, web graphics, and animations. It supports file formats like GIF, PNG, JPEG, SWF (Macromedia Flash), and FHC (Shockwave Freehand).

Video Editing Tools

These tools allow users to edit and manipulate video content.

Adobe Premiere: A professional nonlinear video editing tool. It allows users to arrange video and audio clips in tracks, similar to layers in music production.

Adobe After Effects: A powerful tool for adding visual effects and animations to videos. It supports layers (like in Photoshop) to manipulate objects independently and includes effects such as lighting, shadows, and motion blur.

Final Cut Pro: A professional video editing tool by Apple for Macintosh. It supports video capture, editing, color correction, and exporting to a final video file or live broadcast.

Animation Tools

These tools are used to create animations and 3D models.

Multimedia APIs (Application Programming Interfaces)

APIs provide developers with the tools to create and render animations.

Java3D: A Java-based API for constructing and rendering 3D graphics. It provides basic objects like cubes and splines and works on OpenGL or DirectX for hardware acceleration.

DirectX: A Windows-based API that supports video, images, audio, and 3D animation. It is commonly used in Windows multimedia applications and games.

OpenGL: A widely-used 3D API that runs on multiple platforms, including Windows, Linux, UNIX, and macOS. It is used in gaming and multimedia applications.

Rendering Tools

These tools are used to create 3D models, animations, and special effects.

3D Studio Max: A professional 3D modeling and animation tool used for character animation, game development, and visual effects. It has been used for creating Sony PlayStation game models.

Softimage XSI: Previously called Softimage 3D, this is a high-end tool for animation and special effects in films and games.

Maya: Competes with Softimage and is widely used in 3D modeling and animation, especially for realistic textures, clothes, and fur.

RenderMan: Developed by Pixar, this rendering tool is known for producing high-quality images and surface textures. It has been used in movies like Monsters, Inc. and Final Fantasy: The Spirits Within. It also supports Maya models.

Multimedia Authoring Tools

Multimedia authoring tools allow users to create interactive digital content by integrating text, images, audio, video, and animations. These tools are essential for creating presentations, e-learning content, websites, and interactive applications.

Popular Multimedia Authoring Tools

Adobe Premiere 6 - A video editing software used for cutting, assembling, and enhancing video clips. It allows users to edit, add effects, and create professional-quality videos.

Macromedia Director 8 and MX - A multimedia authoring tool used to create interactive applications, animations, and presentations, often used in game development and educational software.

Flash 5 and MX - A tool for creating animations, interactive content, and web-based multimedia applications. Flash was widely used for online games, videos, and animations before being phased out.

Dreamweaver MX - A web development tool used for designing and coding websites. It supports HTML, CSS, and JavaScript, allowing users to build and manage web pages efficiently.

IMAGE DATA TYPES

Digital images are made up of tiny elements called pixels. Each pixel has a certain amount of information that determines its color or brightness. The way this information is stored defines the image data type. Let's go into more detail about each type.

1-Bit Images (Binary Images)

1-Bit Image /Bit-Map Images

Images consist of *pixels*—picture elements in digital images. A 1-bit image consists of on and off bits only and thus is the simplest type of image. Each pixel is stored as a single bit (0 or 1), the value of the bit indicates whether it is light or dark. Hence, such an image is also referred

to as a *binary image*. It is also sometimes called a 1-bit *monochrome* image, since it contains no color (Figure (3.3)). A 640 x 480 monochrome image requires 37.5 KB of storage.



Figure (3.3): 1-bit (Binary) image

1-bit images can be satisfactory for pictures containing only simple graphics and text. Moreover, fax machines use 1-bit data, so in fact 1-bit images are still important even though storage capacities have increased enough to permit the use of imaging that carries more information.

How It Works:

Each pixel is stored as a single bit:

0 → Black

1 → White

This means it can only represent two colors (black and white), with no shades of gray or color.

Example Calculation:

A 640×480 binary image has $640 \times 480 = 307,200$ pixels.

Since each pixel is 1 bit, the total memory required is:

$307,200 \div 8 = 38.4$ KB (since 8 bits = 1 byte).

Common Uses:

- ✓ Scanned documents
- ✓ Text-based graphics (e.g., QR codes, barcodes)
- ✓ Simple logos

Advantages:

- ✓ Takes up very little storage space.
- ✓ Easy to process in computer vision and printing.

Disadvantages:

- ✗ No color or shading.
- ✗ Cannot store detailed images.

8-Bit Gray-Level Images

Now consider an 8-bit image—that is, one for which each pixel has a *gray value* between 0 and 255 as shown in Figure (3.4). Each pixel is represented by a single byte—for example, a dark pixel might have a value of 10, and a bright one might be 230. The entire image can be thought of as a two-dimensional array of pixel values. We refer to such an array as a *bitmap*—a representation of the graphics/image data that parallels the manner in which it is stored in video memory. A 640 x 480 grayscale image requires over 300 KB of storage.



Figure (3.4): Gray-Scale image

In an **8-bit grayscale image**, each pixel is represented by 8 bits (or 1 byte), which means each pixel can have 256 intensity levels (ranging from 0 to 255). This intensity value is determined by the combination of its **8 bit planes**.

The concept of **bit planes** involves separating the image into its constituent binary layers. Each bit plane corresponds to one bit of the 8-bit pixel value.

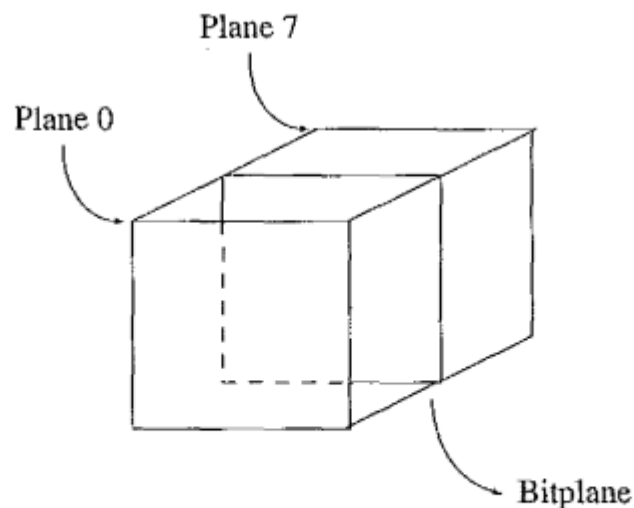


FIGURE 3.2: Bitplanes for 8-bit grayscale image.

24-bit Color Images

In a color 24-bit image, each pixel is represented by three bytes, usually representing RGB. Since each value is in the range 0–255, this format supports $256 \times 256 \times 256$, or a total of 16,777,216, possible combined colors. However, such flexibility does result in a storage penalty: a 640×480 24-bit color image would require 921.6 kB of storage without any compression. An important point to note is that many 24-bit color images are actually stored as 32-bit images, with the extra byte of data for each pixel storing an α (*alpha*) value representing special-effect information. The image forestfire.bmp, in Figure (3.5) a 24-bit image in Microsoft Windows BMP format.

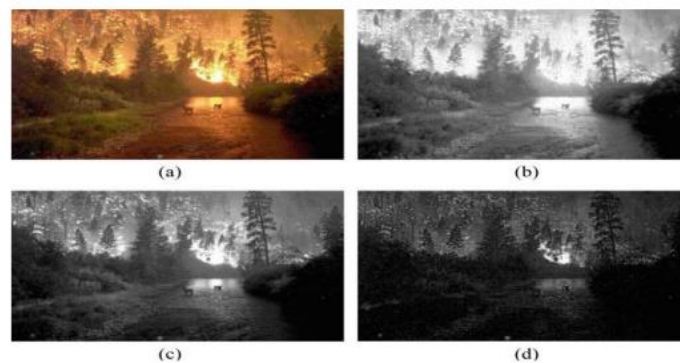


Figure (3.5): High-resolution color and separate R, G, B color channel images. a) original color image; b) c) and d) R, G, and B color channels for this image

8-bit color images

An 8-bit color image is an image where each pixel is represented using 8 bits (1 byte). This allows for 256 different colors in the image.

Instead of using full 24-bit RGB color (where each pixel has 8 bits for red, green, and blue), 8-bit images use a color palette with 256 predefined colors.

Each pixel does not store the full RGB value; instead, it stores an index number (0-255) that points to a specific color in the palette.

Saves space: Since it requires only 1 byte per pixel, it is much smaller in size compared to 24-bit or 32-bit images.

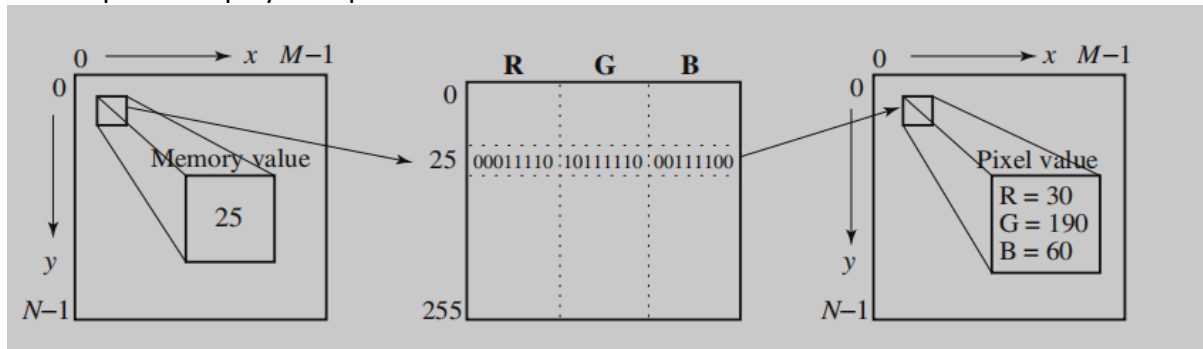
Efficient for graphics: Many systems can display a good range of colors using just 256 colors, making it useful for icons, simple images, and older graphics.

Backward compatibility: Even though modern systems use 24-bit or higher color depths, some still support 8-bit color for compatibility.

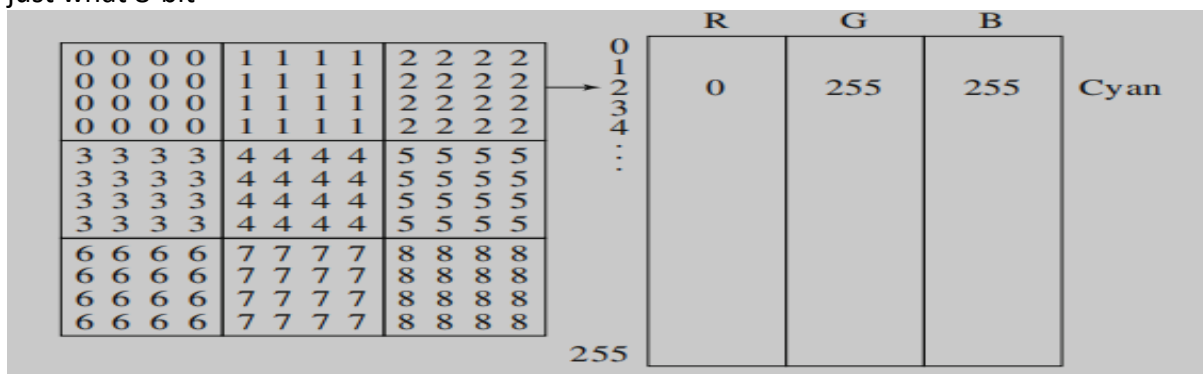
Color Lookup Table

- CLUT is a method used to transform a range of input colors into another range of colors.
- CLUT is a function incorporated in an image processing application.
- Color lookup tables are also used in graphics cards and graphics formats, like GIFs.
- The idea used in 8-bit color images is to store only the index, or code value, for each pixel.

- If a pixel stores, say, the value 25, the meaning is to go to row 25 in a color lookup table (LUT). While images are displayed as two-dimensional arrays of values, they are usually stored in row-column order as simply a long series of values
- The LUT is often called a palette.
- A color picker consists of an array of fairly large blocks of color (or a semicontinuous range of colors) such that a mouse click will select the color indicated. In reality,
- a color picker displays the palette colors associated with index values from 0 to 255.



- While images are displayed as two-dimensional arrays of values, they are usually stored in row-column order as simply a long series of values.
- For an 8-bit image, the image file can store in the file header information just what 8-bit

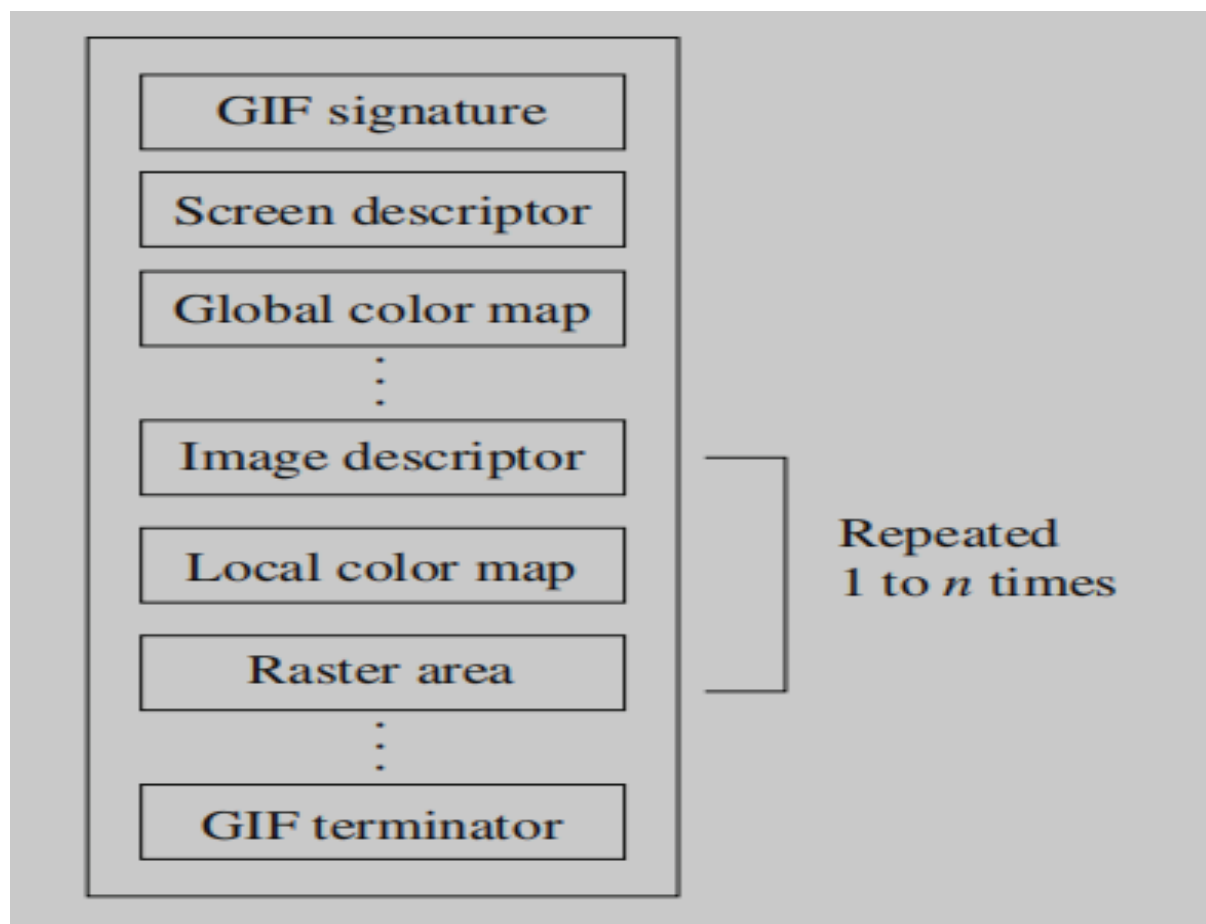


Popular File Formats

GIF

- **Graphics Interchange Format (GIF).**
- It was devised by UNISYS Corporation and Compuserve, initially for transmitting graphical images over phone lines via modems.
- The GIF standard uses the Lempel-Ziv-Welch algorithm modified slightly for image scanline packets to use the line grouping of pixels effectively.
- The GIF standard is limited to 8-bit (256) color images only.
- While this produces acceptable color, it is best suited for images with few distinctive colors (e.g., graphics or drawing).
- GIF is a raster file format designed for relatively basic images that appear mainly on the internet.
- Each file can support up to 8 bits per pixel and can contain 256 indexed colors.
- GIF files also allow images or frames to be combined, creating basic animations.

GIF File Formats



Bits								Byte #	
7	6	5	4	3	2	1	0		
Screen width								1	Raster width in pixels (LSB first)
								2	
Screen height								3	Raster height in pixels (LSB first)
								4	
m	cr		0	pixel				5	
Background								6	Background = color index of screen background (color is defined from the global color map or if none specified, from the default map)
0	0	0	0	0	0	0	0	7	

m = 1 Global color map follows descriptor
 cr + 1 # bits of color resolution
 pixel + 1 # bits/pixel in image

Signature is 6 bytes: GIF87a; the *Screen Descriptor* is a 7-byte set of flags. A GIF87 file can contain more than one image definition, usually to fit on several different parts of the screen. Therefore each image can contain its own color lookup table, a *Local Color Map*, for mapping 8 bits into 24-bit RGB values. However, it need not, and a global color map can instead be defined to take the place of a local table if the latter is not included.

The Screen Descriptor comprises a set of attributes that belong to every image in the file. According to the GIF87 standard, it is defined as in Figure 3.13. *Screen Width* is given in the first 2 bytes. Since some machines invert the order MSB/LSB (most significant byte/least significant byte — i.e., byte order), this order is specified. *Screen Height* is the next 2 bytes. The “m” in byte 5 is 0 if no global color map is given. Color resolution, “cr”, is 3 bits in 0 .. 7. Since this is an old standard meant to operate on a variety of low-end hardware, “cr” is *requesting* this much color resolution.

Bits								Byte #	
7	6	5	4	3	2	1	0		
0	0	1	0	1	1	0	0	1	Image separator character (comma)
Image left								2	Start of image in pixels from the left side of the screen (LSB first)
								3	
Image top								4	Start of image in pixels from the top of the screen (LSB first)
								5	
Image width								6	Width of the image in pixels (LSB first)
								7	
Image height								8	Height of the image in pixels (LSB first)
								9	
m	i	0	0	0	0	0	pixel	10	m = 0 Use global color map, ignore 'pixel' m = 1 Local color map follows, use 'pixel' i = 0 Image formatted in Sequential order i = 1 Image formatted in Interlaced order pixel + 1 # bits per pixel for this image

FIGURE 3.15: GIF image descriptor.

PNG

- **Portable Network Graphics**
- PNG is the only lossless format that web browsers support.
- PNG supports 8 bit, 24 bits, 32 bits and 48 bits data types.
- One version of the format PNG-8 is similar to the GIF format.
- But PNG is the superior to the GIF. It produces smaller files and with more options for colors. It supports partial transparency also.

- One interesting development stemming from the popularity of the Internet is efforts toward more system-independent image formats.
- This standard is meant to supersede the GIF standard and extends it in important ways.
- Special features of PNG files include support for up to 16 bits per pixel in each color channel, i.e., 48-bit color—a large increase.
- Files may also contain gamma-correction information for correct display of color images and α -channel information (up to 16 bits) for such uses as control of transparency.
- It supports both lossless and lossy compression with performance better than GIF.
- PNG is widely supported by various web browsers and imaging software.

Joint Photographic Experts Group(JPEG)

- The most important current standard for image compression is JPEG.
- The JPEG format was developed by the Joint Photographic Experts Group.
- JPEG files are bitmapped images.
- It store information as 24-bit color.
- This is the format of choice for nearly all photograph images on the internet.
- Digital cameras save images in a JPEG format by default.
- It works well on photographs, artwork and similar materials but not so well on lettering, simple cartoons or line drawings. JPEG images work much better than GIFs.

TIFF

- **Tagged Image File Format (TIFF)** is another popular image file format.
- Developed by the Aldus Corporation in the 1980s, it was later supported by Microsoft.
- Its support for attachment of additional information provides a great deal of flexibility.
- The most important tag is a format signifier: what type of compression, etc., is in use in the stored image.
- TIFF can store many different types of images: 1-bit, grayscale, 8-bit, 24-bit RGB, and so on.
- TIFF was originally a lossless format.
- TIFF file format is widely used bitmapped file format.
- It is supported by many image editing applications, software used by scanners and photo retouching programs.

Windows BMP

- BitMap (BMP) is one major system standard image file format for Microsoft Windows.
- It uses raster graphics. BMP supports many pixel formats, including indexed color (up to 8 bits per pixel), and 16, 24, and 32-bit color images.
- It makes use of Run-Length Encoding (RLE) compression and can fairly efficiently compress 24-bit color images due to its 24-bit RLE algorithm.
- BMP images can also be stored uncompressed.

Windows WMF

- Windows MetaFile (WMF) is the native vector file format for the Microsoft Windows operating environment.
- WMF files actually consist of a collection of Graphics Device Interface (GDI) function calls, also native to the Windows environment.
- When a WMF file is “played” (typically using the Windows PlayMetaFile() function) the described graphic is rendered.
- WMF files are ostensibly device-independent and unlimited in size. The later Enhanced Metafile Format Plus Extensions (EMF+) format is device independent.

Netpbm Format

- PPM (Portable PixMap), PGM (Portable GrayMap), and PBM (Portable BitMap) belong to a family of open-source Netpbm formats.
- These formats are mostly common in the linux/unix environments. They are sometimes also collectively known as PNM or PAM (Portable AnyMap).
- These are either ASCII files or raw binary files with an ASCII header for images.

EXIF

- Exchangeable Image File (EXIF) is an image format for digital cameras.
- It enables the recording of image metadata (exposure, light source/flash, white balance, type of scene, etc.) for the standardization of image exchange.
- A variety of tags (many more than in TIFF) is available to facilitate higher quality printing, since information about the camera and picture-taking conditions can be stored and used.
- Ex:- by printers for possible color-correction algorithms.
- It also includes specification of file format for audio that accompanies digital images.

PS

- **PostScript** is a page description language (PDL) used primarily for printing and graphics.
 - It was developed by Adobe Systems in 1982 and became widely used in desktop publishing, graphic design, and printing industries due to its versatility in representing complex documents and images.
 - PostScript is an important language for typesetting, and many high-end printers have a PostScript interpreter built into them.
 - PostScript is a vector-based, rather than pixel based, picture language: page elements are essentially defined in terms of vectors.
- PostScript is a programming language designed to describe the layout of text, images, and graphics on a printed page

Portable Document Format (PDF) file format.

- PDF format is vector graphics with embedded pixel graphics with many compression options.
- When your document is ready to be shared with others or for publication.
- This is only format that is platform independent.
- If you have Adobe Acrobat you can print from any document to a PDF file. From illustrator you can save as .PDF.
- PDF files are designed to be viewed and printed consistently across different platforms, operating systems (Windows, macOS, Linux), and devices (desktop computers, tablets, smartphones).

PTM

- **PTM (Polynomial Texture Mapping)** is a technique for storing a representation of a camera scene that contains information about a set of images taken under a set of lights that each have the same spectrum, but with each placed at a different direction from the scene.
- Polynomial Texture Mapping (PTM) is a technique used in the field of computer graphics to represent a texture map that can capture variations in the appearance of surfaces under different lighting conditions.
- PTM stores a collection of lighting information for each point on a surface, allowing for the reconstruction of how the surface would look under varying lighting angles.

Color Science - Light and Spectra

1. Electromagnetic Waves and Light:

Light is a type of electromagnetic wave, and its color depends on its wavelength. For example:

Shorter wavelengths = Blue/Violet

Longer wavelengths = Red

2. Laser Light:

Laser light has only one specific wavelength, so it appears as a single pure color (e.g., a ruby laser appears red).

3. White Light and Dispersion:

White light (like sunlight) is made up of all colors.

When white light passes through a prism or a diffraction grating, it splits into a rainbow of colors. This is called dispersion.

Similar effects can be seen in rainbows or soap bubbles.

4. Measuring Light:

A device called a spectrophotometer measures light and splits it into its different wavelengths to study its spectrum.

5. Visible Spectrum:

Humans can see light in the range of 400–700 nm (nanometers). This range includes all visible colors, from violet to red.

6. Spectral Power Distribution (SPD):

A graph is used to show the intensity of light across different wavelengths.

The graph helps analyze the properties of light sources, like sunlight or artificial lights.

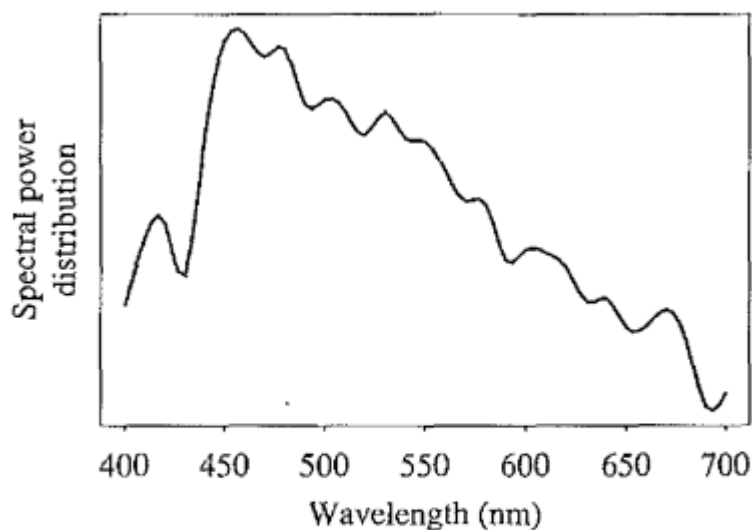


FIGURE 4.2: Spectral power distribution of daylight.

Human Vision

The eye functions like a camera, with the lens focusing images onto the retina.

The retina contains rods and three types of cones:

Rods: Active in low-light conditions, producing grayscale vision.

Cones: Responsible for color vision, with three types sensitive to:

Red (R), Green (G), Blue (B)

The brain processes color information using differences like R–G, G–B, and B–R, as well as an achromatic (black-and-white) channel.

2. Spectral Sensitivity of the Eye

The human eye is most sensitive to light in the middle of the visible spectrum.

The blue receptor (B cone) has a smaller response compared to red (R) and green (G) cones.

Blue evolved late, which is why humans find it a surprising favorite color.

Figure 4.3 shows a graph of cone sensitivities:

The B cone (blue) peaks at shorter wavelengths (~450 nm).

The G cone (green) peaks around 530 nm.

The R cone (red) peaks around 570 nm.

The dashed line $V(\lambda)$ represents the luminous-efficiency function, which shows overall eye sensitivity. It is the sum of response curves to red, green, blue.

Rods are sensitive to a wide range of wavelengths but only contribute to black-and-white vision.

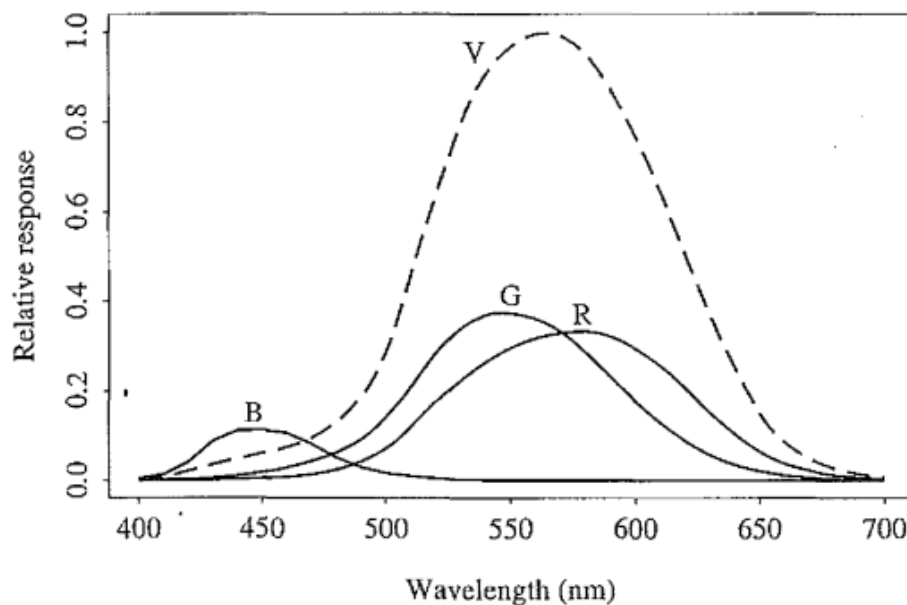


FIGURE 4.3: Cone sensitivities: R, G, and B cones, and luminous-efficiency curve $V(\lambda)$.

Human vision relies on three cone types for color perception and rods for low-light vision.

The eye is most sensitive to green-yellow light, which is why many visual systems optimize for this range. Blue sensitivity is weaker, which aligns with evolutionary trends.

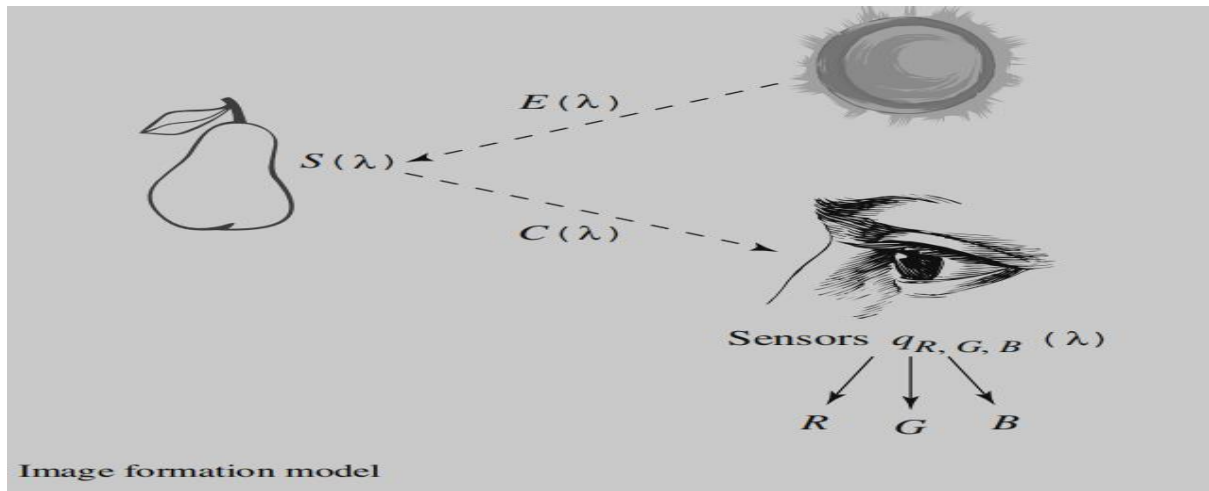
The combination of R, G, and B cone responses allows for full-color vision.

These spectral sensitivity functions are usually denoted by some other letters than R , G , and B , so here let us denote them by the vector function $q(\lambda)$, with components

$$q(\lambda) = [q_R(\lambda), q_G(\lambda), q_B(\lambda)]^T \quad (4.1)$$

$$\begin{aligned}
 R &= \int E(\lambda) q_R(\lambda) d\lambda \\
 G &= \int E(\lambda) q_G(\lambda) d\lambda \\
 B &= \int E(\lambda) q_B(\lambda) d\lambda
 \end{aligned}
 \tag{4.2}$$

Image formation



The image formation situation is thus as follows:

- light from the illuminant with SPD $E(\lambda)$ impinges on a surface, with surface spectral reflectance function $S(\lambda)$, is reflected, and then is filtered by the eye's cone functions $q(\lambda)$.
- The function $C(\lambda)$ is called the color signal and consists of the product of the illuminant $E(\lambda)$ times the reflectance $S(\lambda)$:
 $C(\lambda) = E(\lambda) S(\lambda)$.

$$R = \int E(\lambda) S(\lambda) q_R(\lambda) d\lambda$$

$$G = \int E(\lambda) S(\lambda) q_G(\lambda) d\lambda$$

$$B = \int E(\lambda) S(\lambda) q_B(\lambda) d\lambda$$

Camera Systems

- Now, we humans develop camera systems in a similar fashion, and a good camera has three signals produced at each pixel location (corresponding to a retinal position).
- Analog signals are converted to digital, truncated to integers, and stored.
- If the precision used is 8-bit, then the maximum value for any of R , G , B is 255, and the minimum is 0.

GAMMA CORRECTION

When an image is displayed on a screen (like an old CRT monitor or a TV), the brightness of the light emitted doesn't increase in a straight line with the input signal. Instead, the light output follows a curve, meaning darker areas can appear too dark, and bright areas may not be accurate.

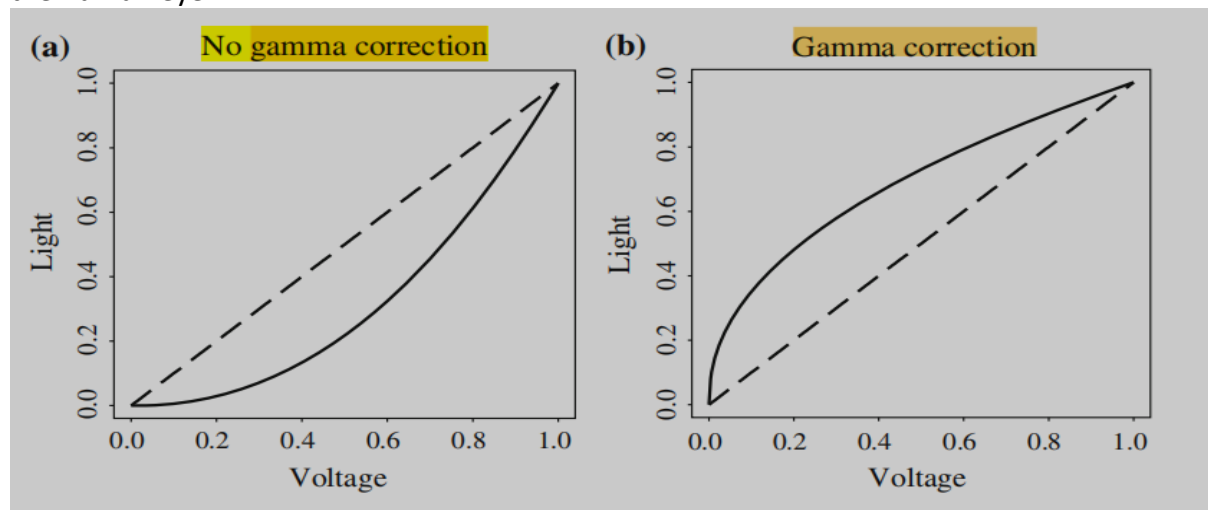
To fix this, a process called gamma correction is applied. This means adjusting the brightness of an image before sending it to the screen so that, after the screen applies its natural curve, the final image appears as intended.

Here's how it works:

Problem: The screen doesn't produce light in a straight-line response to the input signal. Instead, it follows a power curve (gamma), usually around 2.2.

Solution: Before sending the signal to the screen, we apply an inverse gamma correction (raising the signal to the power of $1/\gamma$).

Result: When the screen applies its natural gamma response, the final image looks correct to the human eye



COLOR MATCHING FUNCTIONS

When we try to match or recreate a color, we use a method called color matching. This is important in applications like screen displays, printing, and even dye manufacturing.

How Color Matching Works:

Three Basic Colors (Red, Green, Blue):

Any color can be created by mixing different amounts of three primary colors.

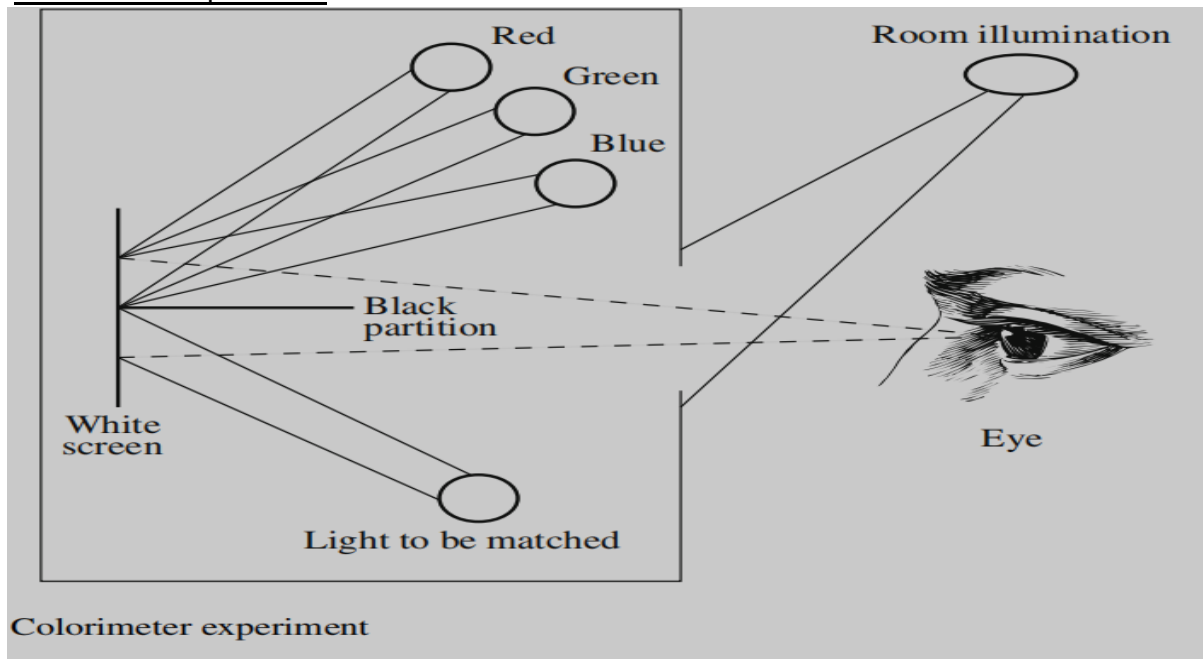
This is why screens and TVs use Red, Green, and Blue (RGB).

Color Matching Experiment:

In early experiments, people were asked to mix red, green, and blue lights to match a target color. This process was used to create a scientific system for defining colors. CIE Standard (Commission Internationale de l'Éclairage):

In 1931, an international organization (CIE) studied color perception and developed color-matching functions. These functions describe how much red, green, and blue are needed to match any visible color.

Colorimeter Experiment

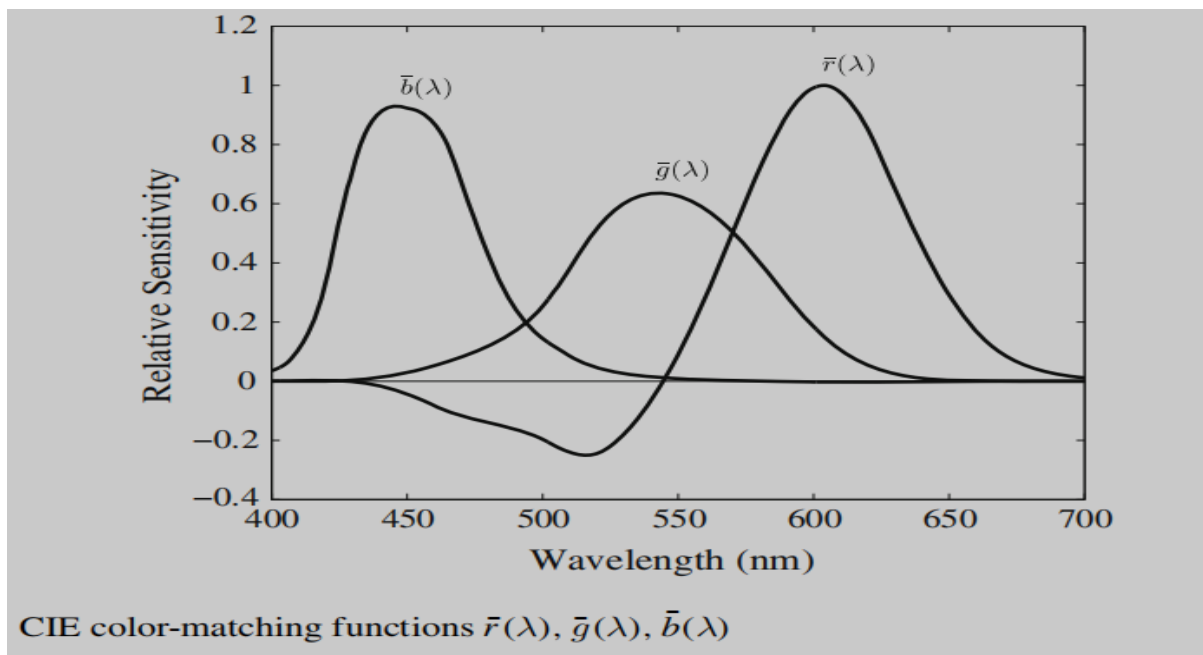


This shows an experiment where a person adjusts red, green, and blue lights to match a target color. A partition blocks direct light, so the observer only sees the final color mix.

(Color-Matching Curves):

This graph shows how much of each primary color (red, green, blue) is needed at different wavelengths of light.

These curves are the foundation of modern color science and are used in displays, photography, and printing.



2. Why Some Curves are Negative

Negative parts in the color-matching functions (e.g.,) occur because not all colors can be represented by the linear combination of the RGB primaries.

To handle such colors, negative lights (subtraction of certain wavelengths) are required to match them.

This implies that these "negative values" indicate adjustments in the color-matching process when comparing certain light spectra to human perception.

CIE Chromaticity Diagram

The CIE Chromaticity Diagram is a tool used in color science to represent colors visually based on their components. This diagram is derived from the color-matching functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$.

The chromaticity diagram simplifies the representation of colors:

- It shows the relationship between different wavelengths and their perceived colors.
- All visible colors can be plotted within this space.

Tristimulus Values (X,Y,Z)

The color of light is described using three primary components, X, Y, and Z, called tristimulus values. These values are calculated using the following integrals:

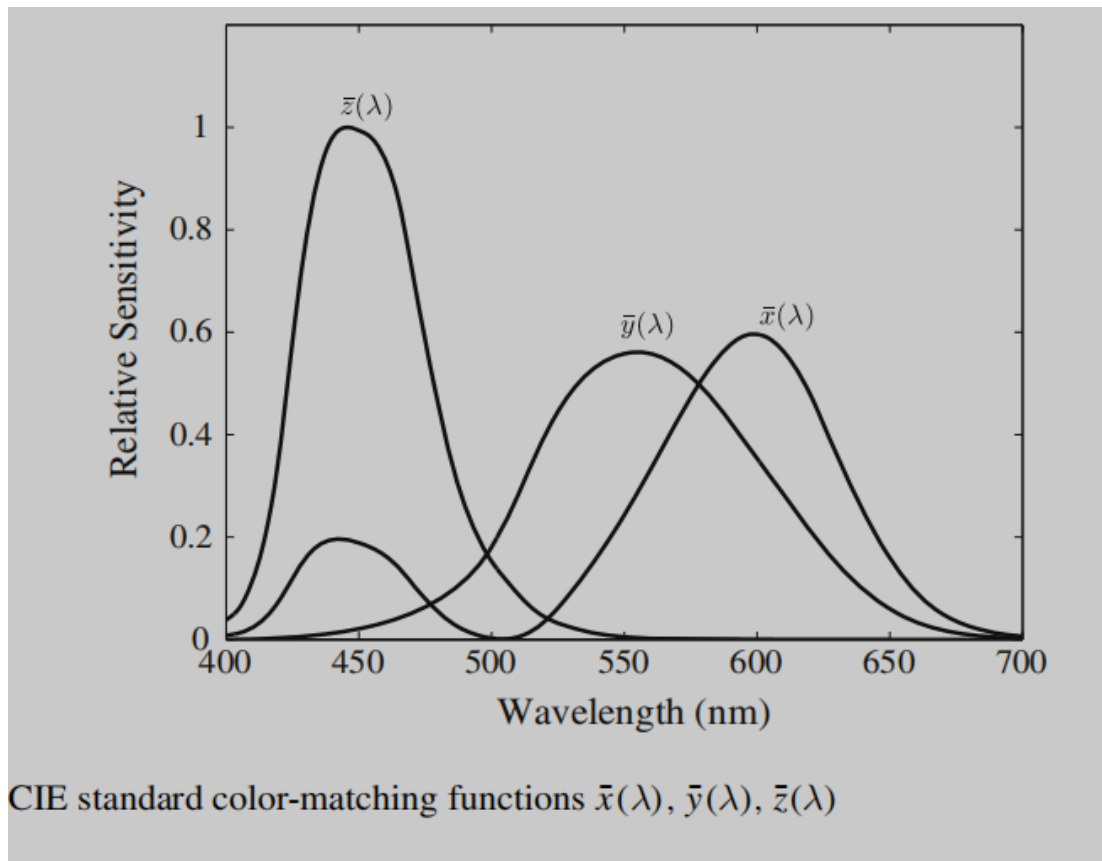
$$\begin{aligned} X &= \int E(\lambda) \bar{x}(\lambda) d\lambda \\ Y &= \int E(\lambda) \bar{y}(\lambda) d\lambda \\ Z &= \int E(\lambda) \bar{z}(\lambda) d\lambda \end{aligned}$$

Where:

$E(\lambda)$: The spectral power distribution (SPD) of the light source. It describes the energy emitted at each wavelength.

$\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$: The CIE standard color-matching functions.

These integrals measure how much of each primary color (red, green, and blue) is required to produce the color of the light source.



1. Luminance (Y):

- This is a measure of the brightness of light, representing the intensity of light as perceived by the human eye.
- It's one of the components in the tristimulus color model (X , Y , Z), where Y specifically encodes brightness.

2. Chromaticity (x , y):

- These are derived from the normalized values of the tristimulus components (X , Y , Z).
- Chromaticity focuses on the color aspects while excluding brightness.
- Equations for chromaticity are given as:

$$x = \frac{X}{X + Y + Z}, \quad y = \frac{Y}{X + Y + Z}, \quad z = \frac{Z}{X + Y + Z}$$

- Here, $x + y + z = 1$, meaning one value (e.g., z) can always be computed as $z = 1 - x - y$.

Values x , y are called chromaticities.

3. Purpose of Chromaticity:

- The normalization process reduces the three-dimensional tristimulus values to two dimensions (x, y) , making it easier to visualize color information.
- This projection forms a 2D chromaticity diagram, which shows the color gamut (range of colors visible to humans).

4. Line of Purples and Spectrum Locus:

- In the chromaticity diagram, the "horseshoe" curve represents all monochromatic light (pure wavelengths).
- The straight line at the bottom (between red and blue) is called the **line of purples**, which consists of colors that cannot be produced by a single wavelength (non-spectral colors).

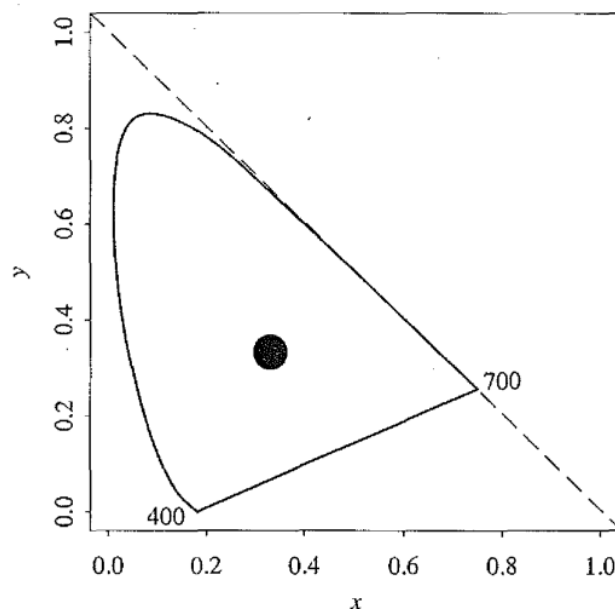


FIGURE 4.11: CIE chromaticity diagram.

Color Models in Images

• RGB COLOR MODEL

- The RGB color model is one of the most widely used color representation method in computer graphics.
- It use a color coordinate system with three primary colors:
R(red), G(green), B(blue)
- Each primary color can take an intensity value ranging from 0(lowest) to 1(highest).
- Mixing these three primary colors at different intensity levels produces a variety of colors.

- The collection of all the colors obtained by such a linear combination of red, green and blue forms the cube shaped RGB color space.
- **Additive Color model.**
- Each primary color can take an intensity value ranging from 0(lowest) to 1(highest).
- Mixing these three primary colors at different intensity levels produces a variety of colors.
- The collection of all the colors obtained by such a linear combination of red, green and blue forms the cube shaped RGB color space.
- Today, the additive color model is primarily used to visualize, represent and display images in electronic systems, such as TVs, computer monitors and mobile phones.

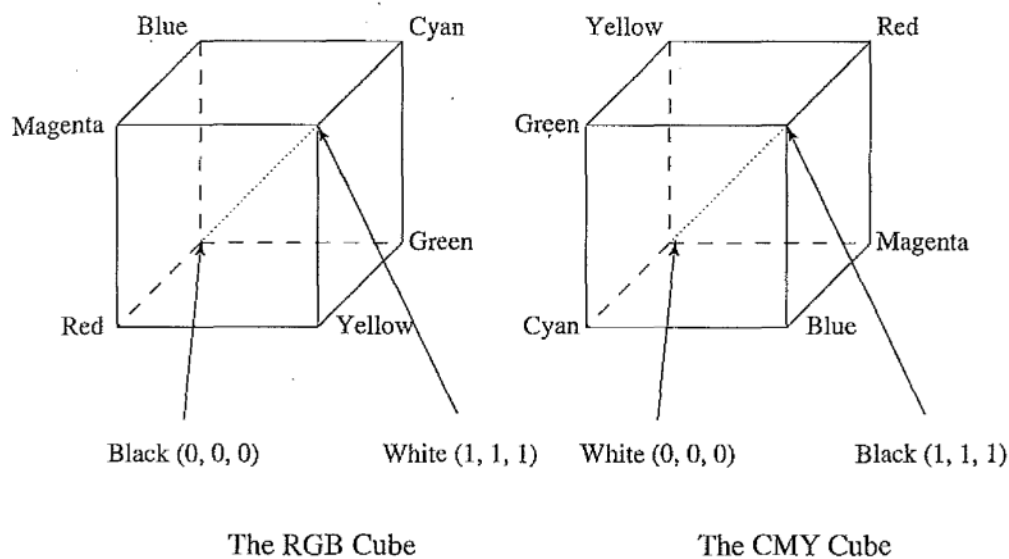
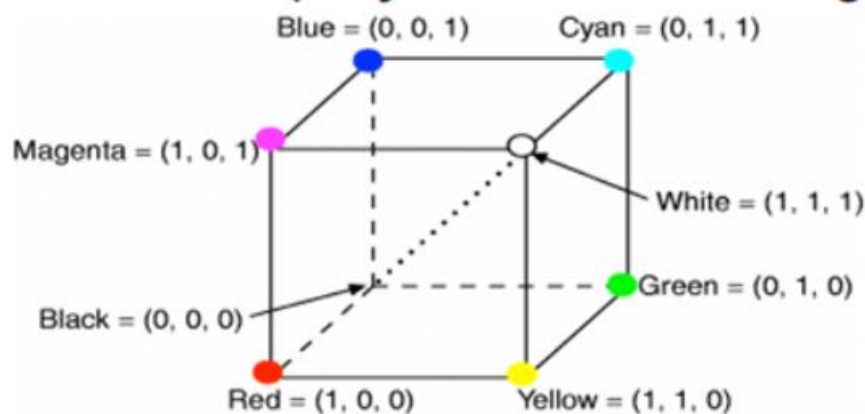


FIGURE 4.15: RGB and CMY color cubes. (This figure also appears in the color insert section.)

The RGB cube (Grays on dotted main diagonal)



• CMY Color Model

The **CMY model** is based on **subtractive color mixing**, used in printing and pigment-based systems. Subtractive mixing occurs by **removing (absorbing)** wavelengths of light.

- The CMY color model use a **subtraction process** and this concept is used in the **printer**.
- In CMY model, we begin with white and take away the appropriate primary components to yield a desired color.
- **Example:**
If we subtract red from white, what remains consists of green and blue which is cyan.
- In the subtractive color model, pigment is used to produce color using reflected light.
- This color model is used in printing, silk-screening, painting and other mediums that add pigment to a substrate.
- The **CMY color model** is a **subtractive color model** used in color printing and photography. It is based on the three primary colors of pigment:
 - Cyan (C)
 - Magenta (M)
 - Yellow (Y)

How the CMY Model Works

- Unlike the RGB model, which is additive (colors are created by adding light), the CMY model is subtractive—it works by absorbing certain wavelengths of light and reflecting others.
- Cyan absorbs red and reflects green and blue.
- Magenta absorbs green and reflects red and blue.
- Yellow absorbs blue and reflects red and green.
- When all three CMY colors are combined at full intensity, they ideally produce black (absorbing all light). However, due to ink impurities, a separate black (K) ink is added in practical printing, forming the CMYK model.

• CMY to RGB Conversion

- Mathematically, CMY can be derived from RGB using the formula:
- This means each CMY color is the **inverse** of its corresponding RGB counterpart.

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Then the inverse transform is

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$

• Undercolor Removal :CMYK System

The **CMYK color model** is an extension of the **CMY (Cyan, Magenta, Yellow) color model**, used mainly in **printing**. The "**K**" in **CMYK stands for Black (Key)** because black ink is used in addition to CMY to produce darker tones and sharper prints.

Why is Black (K) Needed?

In an ideal **CMY model**, combining **100% of Cyan, Magenta, and Yellow** should create **pure black**. However, in real-world printing, the inks are not perfect, and their combination produces a **muddy brownish-black** instead of true black.

To fix this, **black ink (K)** is added separately. This process is called **Undercolor Removal (UCR)**, where a part of the CMY mix that would form black is removed, and pure black ink is used instead.

Formula for Black (K) Component:

$$K = \min(C, M, Y)$$

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} \Rightarrow \begin{bmatrix} C - K \\ M - K \\ Y - K \end{bmatrix}$$

Printer Gamut

- A printer gamut refers to the range of colors a printer can reproduce using CMY (Cyan, Magenta, Yellow) or CMYK (Cyan, Magenta, Yellow, Black) inks.
- The gamut is smaller than what a computer screen (RGB) can display because printers use subtractive color mixing.

2. How Do Printers Create Colors?

- Printers lay down transparent layers of ink on a generally white substrate (paper).
- The colors are formed by controlling the density of CMY inks.
- The center of the gamut corresponds to black, where all three inks overlap.
- The edges represent pure CMY colors.

3. Comparison with RGB Gamut

- The printer gamut is smaller than the NTSC color gamut (which is a standard for television screens).
- This means some colors that you see on a screen cannot be printed exactly.

4. Optical Density and Color Formation

- The optical density (D) is used to measure how much ink is absorbed or reflected.
- A logarithmic relationship exists between ink density and light transmission.
- This helps in defining accurate color reproduction.

•

Digital Video

Digital video refers to video content that is recorded, stored, and played back using digital formats instead of traditional analog methods like VHS tapes. It is widely used in TV broadcasts, online streaming, video editing, and filmmaking.

Advantages of Digital Video

- ✓ Higher quality compared to analog.
- ✓ Easy to edit & share (supports software-based editing).
- ✓ Efficient storage & transmission (compressed formats save space).
- ✓ Compatible with modern devices (smartphones, computers, TVs).

Chroma Subsampling

Chroma subsampling is a technique used in digital video and image compression to reduce file size without significantly affecting perceived quality. It works by reducing the amount of color information while keeping the brightness (luminance) details intact. This is based on the fact that the human eye is more sensitive to brightness than color.

How It Works

4:4:4 (No Subsampling): Every pixel has full color and brightness information. This provides the highest quality but takes up the most storage space.

4:2:2 (Half Horizontal Color Subsampling): The brightness (Y) is stored for every pixel, but color information (Cb and Cr) is stored for every two pixels instead of every one. This reduces data while maintaining decent quality.

4:2:0 (Both Horizontal and Vertical Subsampling): The color information is stored for every four pixels, meaning both horizontal and vertical resolutions of color are reduced. This saves even more data while still looking good to the human eye.

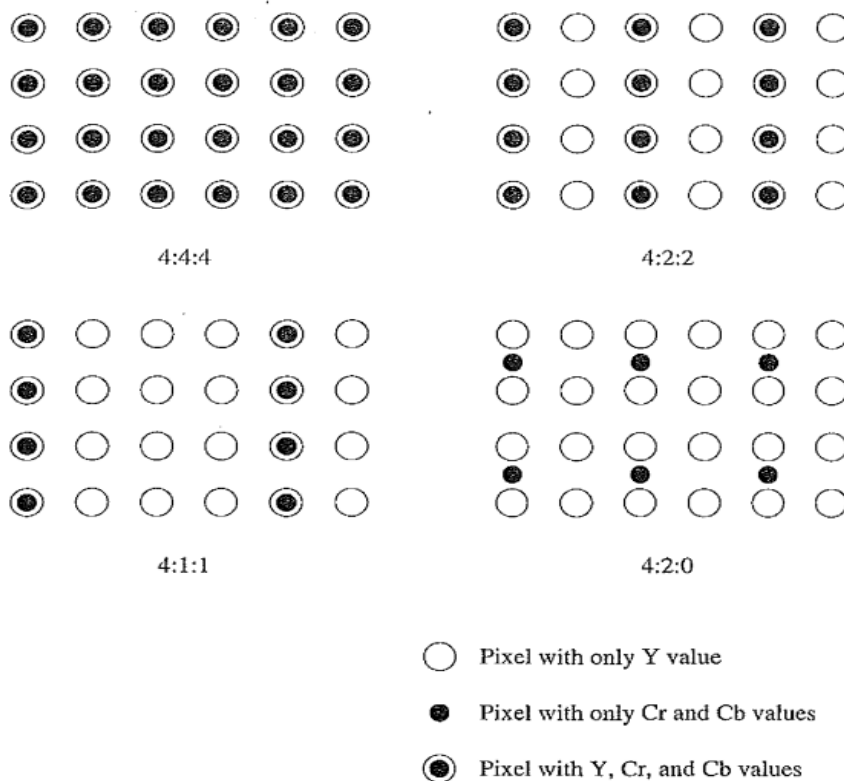
4:1:1 (More Aggressive Subsampling): Color information is reduced even further, which can lead to noticeable loss in quality.

Why Use Chroma Subsampling?

Saves storage and bandwidth

Works well for videos and images where slight color loss is not easily noticed

Used in formats like JPEG (for images) and MPEG (for videos) to balance quality and efficiency.



CCIR Standards for Digital Video

The CCIR (Consultative Committee for International Radio) created standards for digital video to ensure consistency in how videos are recorded and transmitted. One of the most important standards they developed is CCIR-601, which later became ITU-R 601, a widely used standard for professional video applications, including DV (Digital Video) formats.

NTSC Version of CCIR-601

Resolution: The NTSC version of this standard has 525 scan lines, each containing 858 pixels. However, only 720 pixels are visible because some are part of the "blanking period" (extra data used for synchronization).

Chroma Subsampling: NTSC uses 4:2:2 chroma subsampling, meaning it retains more color information compared to 4:2:0.

Data Rate: Each pixel is stored using two bytes (8 bits for brightness "Y" and 8 bits alternating between color components "Cb" and "Cr"). This results in a total data rate of approximately 216 Mbps (megabits per second).

Interlaced Scan & Extra Data Capacity

The CCIR 601 standard uses interlaced scanning, meaning each frame is split into two fields, one containing even lines and the other containing odd lines. This helps reduce flickering on older display technologies.

Some of the blank space in the signal can be used to carry extra data, such as audio, translations, or error-correction information.

High Definition TV

HDTV provides a significantly higher resolution and better image quality than conventional analog TV.

It was developed to enhance the viewing experience, particularly for larger screens.

Historical Development:

Early HDTV developments began in Japan in the 1980s with NHK's system.

The U.S. and other countries later developed digital HDTV standards to improve efficiency and compatibility.

The Advanced Television Systems Committee (ATSC) established the HDTV standards in the 1990s.

Technical Features of HDTV:

Uses digital broadcasting, unlike traditional NTSC and PAL analog systems.

Offers resolutions of 720p, 1080i, and 1080p, which significantly improve picture clarity.

Supports MPEG-2 compression for efficient transmission.

Uses 16:9 aspect ratio, providing a wider screen compared to the traditional 4:3 format.

Comparison with Conventional TV:

HDTV has better luminance resolution and color accuracy than standard TV formats like NTSC and PAL.

Conventional TVs typically have interlaced scanning, whereas HDTV can have progressive scanning (e.g., 720p).

Bandwidth Considerations:

HDTV requires over 20 MHz bandwidth, which does not fit within the existing 6 MHz or 8 MHz TV channels.

Compression techniques are needed, and HDTV may use multiple channels for signal transmission.

FCC Decision & HDTV Development:

In 1987, the FCC (Federal Communications Commission) required HDTV to be compatible with existing NTSC standards.

By 1990, the FCC changed its stance, advocating full-resolution digital HDTV, leading to its simultaneous broadcast alongside NTSC.

In 1993, a "grand alliance" was formed by companies like General Instruments, MIT, Zenith, AT&T, Thomson, Philips, and Sarnoff, leading to the creation of the ATSC standard in 1995.

Digital Scanning Formats & Compression:

Frame rates follow NTSC standards, including 60.00 Hz, 59.94 Hz, 30.00 Hz, 29.97 Hz, 24.00 Hz, and 23.98 Hz.

MPEG-2 is used for video compression, and AC-3 (Dolby 5.1 surround sound) is used for audio.

TABLE 5.3: Digital video specifications.

	CCIR 601 525/60 NTSC	CCIR 601 625/50 PAL/SECAM	CIF	QCIF
Luminance resolution	720 × 480	720 × 576	352 × 288	176 × 144
Chrominance resolution	360 × 480	360 × 576	176 × 144	88 × 72
Color subsampling	4:2:2	4:2:2	4:2:0	4:2:0
Aspect ratio	4:3	4:3	4:3	4:3
Fields/sec	60	50	30	30
Interlaced	Yes	Yes	No	No